Hokyun Lee

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Education

Seoul National University — Seoul, South Korea
Ph.D. candidate in Department of Intelligence and Information, Advisor: Jaeheung Park — 2023–Current

Seoul National University — Seoul, South Korea
A combined M.S./Ph.D. program in Department of Intelligence and Information. — 2021–Current

Daegu Gyeongbuk Institute of Science and Technology — Dageu, South Korea
B.E. in in the School of Undergraduate Studies — 2017–2021

Experience

Seoul National University/DYROS Lab. — Seoul, South Korea
System Engineer/Dynamic Robotic Systems Labratory — 2020–2022

- ANA AVATAR XPRIZE COMPETITION
- Developing the TOCABI avatar system integrating control, haptics, sensing, and design.

Daegu Gyeongbuk Institute of Science and Technology — Dageu, South Korea
Control Engineer/Bio Robotics and Mechatronics Laboratory, Advisor: Dongwon Yun — 2020

- Development of a Quadruped Walking Robot for Campus Safety at DGIST
- A quadruped robot for campus safety at DGIST through autonomous patrol, hazard detection.

Daegu Gyeongbuk Institute of Science and Technology — Dageu, South Korea
System Engineer/Motion Control Laboratory, Advisor: Sehoon Oh — 2019

- Research on the Development of a High-Speed Quadruped Robot
- Minimal-Impact Locomotion of a Robotic Leg Using a Modular Series Elastic Actuator

Publications

[1] J. Ahn, J. Jung, Y. Lee, **H. Lee**, S. Haddadin, and J. Park, "Efficient computation of whole-body control utilizing simplified whole-body dynamics via centroidal dynamics", *International Journal of Control, Automation and Systems*, vol. 23, no. 9, pp. 2656–2665, 2025.

[2] D. Kim, **H. Lee**, J. Cha, and J. Park, "Bridging the reality gap: Analyzing sim-to-real transfer techniques for reinforcement learning in humanoid bipedal locomotion", *IEEE Robotics & Automation Magazine*, 2025.

[3] J. Kim, **H. Lee**, and J. Park, "Real-time whole-body model predictive control for bipedal locomotion with a novel kino-dynamic model and warm-start method", *arXiv preprint arXiv:2505.19540*, 2025.

[4] J. Ahn, J. Jung, **H. Lee**, and J. Park, "Time-efficient contact consistent whole-body control framework via reduced-dimension dynamics construction", *IEEE Access*, 2024.

[5] **H. Lee**, G. Park, J. Shin, B. Park, and J. Park, "Foot-operated telelocomotion interface for avatar robots utilizing mecanum wheel-based mobile platforms", in *2023 23rd International Conference on Control, Automation and Systems (ICCAS)*, IEEE, 2023, pp. 651–656.

Teaching

- **Head Teaching Assistant** at Seoul National University 2023–Current
  DYROS Robotics Boot Camp II (Robot Design, SI, Simulation and Control)
- **Teaching Assistant** at Seoul National University 2022–2023
  DYROS Robotics Boot Camp II (Robot Design, SI, Simulation and Control)
- **Teaching Assistant** at Seoul National University 2022–Current
  DYROS Robotics Boot Camp I (ROS: Robot Operating System)

Skills

- **Software:** C++, Python, MATLAB, PyTorch, Numpy, Anaconda, Docker, OpenCV, C, R
- **Hardware:** Fusion360, 3D Printing, Wiring, Soldering
- **System Integration:** RTLinux, ROS, EtherCAT, Shared Memory, Multi-threading , Qt, RPi, Jetson
- **Applications:** Isaac Gym/Lab, MuJoCo, CoppeliaSim, MoveIt, Git, VS code, Discord, Notion, Slack

Languages

- **Language:** Korean (mother tongue)
- **Language:** English (intermediate)
- **Language:** Chinese (elementary)
- **Language:** Japanese (elementary)